ChE 210 HW #8

43/53

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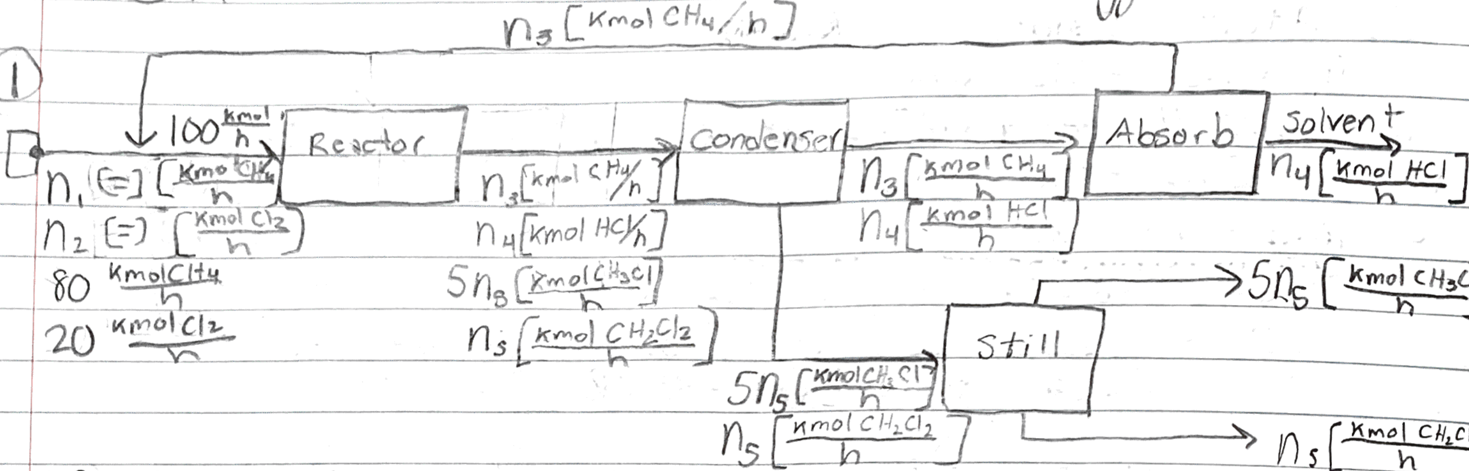
(1)



R_1: $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (ξ_1)

R_2: $CH_3Cl + Cl_2 \longrightarrow CH_2Cl_2 + HCl$ (ξ_2)

(a) DOF $(4a+4b)-(5a+5b+5c+5d)$

Overall $=(16+2)-(1+1+1+15)=0$, specified

Reactor $=(6+2)-(1+1+1+5)=0$, specified

Condenser $=(8+0)-(0+0+0+4)=4$, underspecified

Absorber $=(4+0)-(0+0+0+2)=2$, underspecified

Still $=(4+0)-(0+0+0+2)=2$, underspecified

Mixing Point $=(5+0)-(1+1+0+2)=1$, underspecified

(b) Species Balance on Mixing Point

CH_4: $\dot{n}_1+\dot{n}_3 = 80 \frac{\text{kmol } CH_4}{h} \Rightarrow \dot{n}_1 = 80-\dot{n}_3, \quad \dot{n}_3 = 80-\dot{n}_1$

Cl_2: $\dot{n}_2 = 20 \frac{\text{kmol } Cl_2}{h}$

Atomic Balance on Reactor

C: $80(1) = (1)\dot{n}_3 + (1)5\dot{n}_5 + (1)\dot{n}_5 \Rightarrow \dot{n}_3 + 6\dot{n}_5 = 80 \frac{kmol}{h}$

H: $(4)\dot{n}_3 + \dot{n}_4 + (17)\dot{n}_5 = (4)80 \frac{kmol}{h}$

Cl: $\dot{n}_4 + (7)\dot{n}_5 = (2)20 \frac{kmol}{h}$

Solving for $\dot{n}_3, \dot{n}_4, \dot{n}_5$

$$\left[\begin{array}{cccc|c} 1 & 0 & 6 & -80 & 0 \\ 4 & 1 & 17 & -320 & 0 \\ 0 & 1 & 7 & -40 & 0 \end{array}\right] \xrightarrow[\text{[calc]}]{\text{rref(}} \left[\begin{array}{ccccc} 1 & 0 & 0 & -62.9 & 0 \\ 0 & 1 & 0 & -20 & 0 \\ 0 & 0 & 1 & -2.86 & 0 \end{array}\right]$$

$$\boxed{\begin{aligned} \dot{n}_3 &= 62.9 \tfrac{kmol}{h} \\ \dot{n}_4 &= 20 \tfrac{kmol}{h} \\ \dot{n}_5 &= 2.86 \tfrac{kmol}{h} \end{aligned}}$$

(c) Atomic Balance on Reactor

C: $\dot{n}_1 + \dot{n}_3 = \dot{n}_3 + 6\dot{n}_5$

H: $4\dot{n}_1 = 4\dot{n}_3 + \dot{n}_4 + 17\dot{n}_5$

$\Rightarrow 17\dot{n}_5 = 4\dot{n}_1 - 4\dot{n}_3 - \dot{n}_4$

Cl: $2\dot{n}_2 = \dot{n}_4 + 7\dot{n}_5$

$\Rightarrow 7\dot{n}_5 = 2\dot{n}_2 - \dot{n}_4$

$$5\dot{n}_5 = \frac{\dot{m}_5}{M_{CH_3Cl}} = \frac{1000 \text{ kg/h}}{(12.01+(1.008)3+35.45)} = 19.8 \frac{\text{kmol}}{h}$$

$5\dot{n}_5 = 19.8 \frac{\text{kmol}}{h}$

$\Rightarrow \boxed{\dot{n}_5 = 3.96 \frac{\text{kmol}}{h}}$ (plug back in)

$67.32 \frac{\text{kmol}}{h} = 4\dot{n}_1 - 4\dot{n}_3 - \dot{n}_4$

$\Rightarrow \boxed{\dot{n}_1 = 23.76 \frac{\text{kmol}}{h}}$

$27.72 \frac{\text{kmol}}{h} = 2\dot{n}_2 - \dot{n}_4$

$\dot{n}_1 + \dot{n}_3 = 4\dot{n}_2$ (molar composition of feed into reactor)

(Using Excel)

$$\boxed{\begin{aligned} \dot{n}_1 &= 23.76 \text{ kmol/h} \\ \dot{n}_2 &= 8.36 \text{ kmol/h} \\ \dot{n}_3 &= 9.68 \text{ kmol/h} \end{aligned}}$$

$\dot{n}_{feed} = \dot{n}_1 + \dot{n}_2 = 32.13 \text{ kmol/hr}$

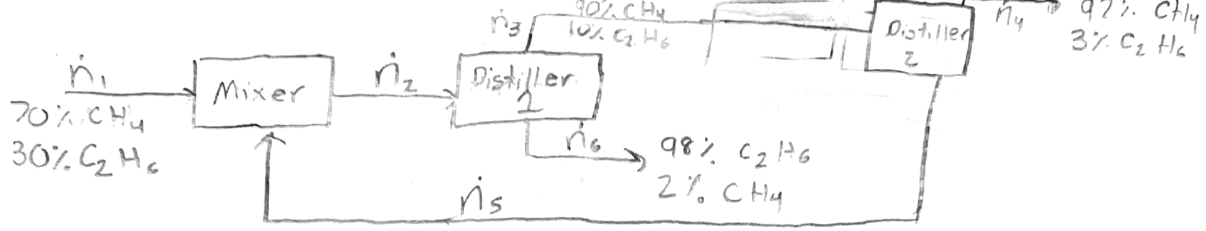
$$x_{CH_4,feed} = \frac{23.76 \text{ kmol/h}}{32.13 \text{ kmol/h}} = 74\%$$

$x_{Cl_2,feed} = 100\% - 74\% = 26\%$

For Desired Product:

$\dot{n}_{feed} = 32.13 \text{ kmol/h}$, 74% CH_4 & 26% Cl_2

$\dot{n}_{recycle} = \dot{n}_3 = 9.68 \text{ kmol/h}$, 100% CH_4 only.



(2) Murphy 5.32

Basis: 100 gmol/min

Species Balance on Mixer

CH_4: $70 + X_{CH_4,i}(100) = X_{CH_4,F}(100)$

C_2H_6: $30 + X_{C_2H_6,i}(100) = X_{C_2H_6,F}(100)$

$X_{CH_4} + X_{C_2H_6} = 1$

on Distiller 1

$X_{CH_4,F}(100) = 0.9\dot{n}_2 + 0.02\dot{n}_3$

$X_{C_2H_6,F}(100) = 0.1\dot{n}_2 + 0.98\dot{n}_3$

On Distiller 2

CH_4: $0.9\dot{n}_2 = 0.97\dot{n}_4 + X_{CH_4}(100)$

C_2H_6: $0.1\dot{n}_2 = 0.03\dot{n}_4 + X_{C_2H_6}(100)$

$X_{C_2H_6} + X_{CH_4} = 1$

Overall Balance

$(70 = 0.97\dot{n}_4 + 0.02\dot{n}_3)\cdot 49$

$30 = 0.03\dot{n}_4 + 0.98\dot{n}_3$

$\Rightarrow$

$3430 = 47.53\dot{n}_4 + \cancel{0.98\dot{n}_3}$

$-\ 30 = 0.03\dot{n}_4 + \cancel{0.98\dot{n}_3}$

$3400 = 47.50\dot{n}_4$

Final Product $\rightarrow$ $\boxed{\dot{n}_4 = 71.58 \text{ gmol/min}}$

$70 = 0.97(71.58) + 0.02\dot{n}_3$

$\boxed{\dot{n}_3 = 28.37 \text{ gmol/min}}$

Total mass for Distiller 1: $\dot{n}_F = \dot{n}_3 + \dot{n}_2$

$200 = 28.32 + \dot{n}_2 \Rightarrow \boxed{\dot{n}_2 = 171.63 \text{ gmol/min}}$

$\dot{n}_2$

CH_4: $0.9(171.63) = 154.467$ gmol CH_4/min

C_2H_6 $0.1(171.63) = 17.163$ gmol C_2H_6/min

Species Balance on Distiller 2

CH_4: $154.467 = 0.97(71.58) + X_{CH_3}(100)$

$\boxed{X_{CH_4} = 85\%}$ recycled stream

$\boxed{X_{C_2H_6} = 15\%}$ Composition

$\dot{n}_F$ Species Balance

CH_4: $0.85(100) + 70 = X_{CH_4}(200)$

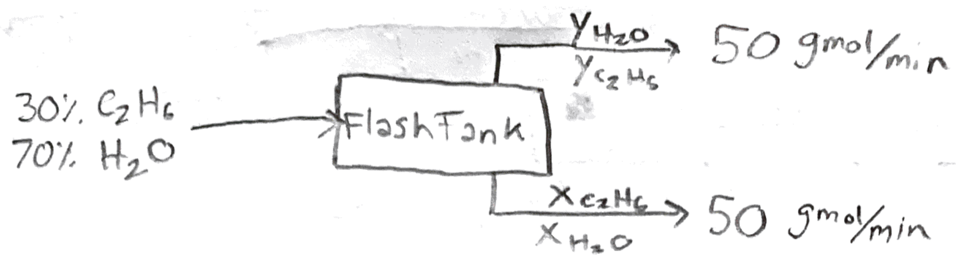
Feed Stream $\boxed{X_{CH_4} = 0.775}$

Composition $\boxed{X_{C_2H_6} = 0.225}$

Fractional Recoveries

CH_4: $\dfrac{X_{CH_4,4}\dot{n}_4}{X_{CH_4,1}\dot{n}_1} = \dfrac{(0.97)(71.58)}{0.7(100)} = 0.9919$ or 99.2%

C_2H_6: $\dfrac{X_{C_2H_6,3}\dot{n}_3}{X_{C_2H_6,1}\dot{n}_1} = \dfrac{(0.98)(28.37)}{(0.3)(100)} = 0.9267$ or 92.67%



③ Murphy 5.34

Basis: 100 gmol/min

$0.3(100) = 30 = Y_{C_2H_6}(50) + X_{C_2H_6}(50)$ $\qquad Y_{C_2H_6} + Y_{H_2O} = 1$

$0.7(100) = 70 = Y_{H_2O}(50) + X_{H_2O}(50)$ $\qquad X_{C_2H_6} + X_{H_2O} = 1$

Test Temps from Table B.11

@ $T = \cancel{81.5}\,°C$ $\qquad Y_{C_2H_6} = 0.5826,\ X_{C_2H_6} = 0.3273$

$0.5826(50) + 0.3273(50) \neq 30$ (need smaller values)

@ $T = 84.1°C$ $\qquad Y_{C_2H_6} = 0.5089,\ X_{C_2H_6} = 0.1661$

$0.5089(50) + 0.1661(50) = 35.75 \neq 30$ (Too high)

$$\boxed{\begin{array}{ll} @\,T = 85.3°C & 30 = 0.4704(50) + 0.1238(50) \\ & 30 \approx 29.71 \end{array}}$$

$$f_{R,C_2H_6(v)} = \frac{Y_{C_2H_6}\,\dot{n}_{vapor}}{Z_{C_2H_6,F}\,\dot{n}_F} = \frac{0.47(50)}{0.30(100)} = 0.78\bar{3}$$

$$f_{R,H_2O(l)} = \frac{X_{H_2O}\,\dot{n}_{liquid}}{Z_{H_2O,F}\,\dot{n}_F} = \frac{0.876(50)}{0.70(100)} = 0.626$$

$$\alpha_{C_2H_6-H_2O} = \frac{Y_{C_2H_6}}{X_{C_2H_6}} \cdot \frac{Y_{H_2O}}{X_{H_2O}} = \left(\frac{0.47}{0.124}\right)\left(\frac{0.876}{0.53}\right) = \boxed{6.26}$$

Comment Summary

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1. -4. Recalculate n2 and n3